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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 19

Malicious Software

Malware

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CHAPTER 19

Malicious Software (Malware)

Computer Awareness

IN THIS CHAPTER

- Viruses, worms and Trojans
- Spyware and ransomware
- RATs and rootkits
- Web attacks
- Cyberwarfare
- Information Technology Act

MALICIOUS SOFTWARE (MALWARE)

Malware (Malicious Software) refers to any software intentionally designed to damage, disrupt, steal data from, or gain unauthorized access to a computer system without the user's knowledge or permission.

Objectives of Malware

- Steal Information
- Damage Files
- Spy on Users
- Gain Unauthorized Access
- Disrupt System Operations

Exam Point

- Malware = Malicious Software
- Malware is designed to harm computers or networks.

HISTORY OF COMPUTER VIRUS

Creeper Virus

The world's first computer virus-like program was Creeper, created by Bob Thomas (Robert Thomas) in 1971.

Characteristics

- Self-replicating program
- Moved from one computer to another
- Displayed the message:

"I'M THE CREEPER, CATCH ME IF YOU CAN!"

Exam Point

- First Virus-like Program = Creeper
- Created by Bob Thomas

Reaper

The world's first antivirus program was Reaper.

Function

- Designed to detect and remove the Creeper virus.

Exam Point

- Early Creeper-removal program = Reaper

TYPES OF MALWARE

Malware

- Virus
- Worm
- Trojan Horse
- Spyware
- Ransomware
- Adware
- Rootkit
- Boot Sector Virus

1. VIRUS

A virus is a type of harmful software that is designed to damage or disrupt computer systems. Unlike standalone programs, it attaches itself to a legitimate file or application. When the infected file or program is run, the virus becomes active, often spreading to other files and systems. It can be created with the intent to steal data, corrupt files, slow down the system, or cause other forms of harm.

Characteristics

- Requires a host file
- Self-replicates
- Can corrupt or delete data
- Spreads through infected files

Example

File infecting virus.

Exam Point

- Virus needs a host file to spread.

2. WORM

A worm is self-replicating malware that can spread independently, often through networks. Unlike a file-infecting virus, it does not require a host file.

Exam Point

- Worm does not require a host file.

3. TROJAN HORSE

A Trojan Horse is a type of harmful software that pretends to be a safe or useful program. It tricks users by looking like something they want to install, such as a game, app, or update. Once the user runs the program, the Trojan secretly carries out harmful actions, like stealing personal information, damaging files, or allowing hackers to access the computer. Unlike viruses, Trojans do not spread by themselves; they rely on the user to download and install them. Being careful about what you download and using antivirus software can help protect against Trojan Horses.

Characteristics

- Appears useful
- Performs hidden malicious actions
- Does not self-replicate

Example :Fake antivirus software

Exam Point

- Trojan tricks users into installing it.

VIRUS vs WORM vs TROJAN

Feature	Virus	Worm	Trojan
Self Replication	Yes	Yes	No
Host File Required	Yes	No	No
User Action Needed	Usually	No	Yes
Disguised as Genuine Software	No	No	Yes
Network Spread	Limited	Very Fast	Limited

4. SPYWARE

Spyware is a type of software that secretly watches what a user is doing on a computer or phone and collects information without asking for permission.

Information Collected

- Passwords
- Browsing History
- Banking Information
- Personal Data

Exam Point

- Spyware = Secret Monitoring Software

5. BOOT SECTOR VIRUS

A virus that infects the boot sector of storage devices.

Effect

- Prevents system startup
- Corrupts boot process

Exam Point

- Targets the Master Boot Record (MBR).

6. RANSOMWARE

Ransomware is a type of harmful software that cybercriminals use to lock your files or computer. It works by secretly entering your computer and encrypting (locking) your important files so that you cannot open them anymore. Once the files are locked, the attacker shows a message asking you to pay money, called a ransom, to get the key that can unlock your files.

It is called “ransomware” because the cybercriminals are asking for ransom, just like in a kidnapping, but here your data is kidnapped. Paying the ransom is risky because there is no guarantee that the attackers will really unlock your files.

Example :WannaCry Ransomware

Exam Point

- Ransomware locks files and demands payment.

7. ADWARE

Software that automatically displays unwanted advertisements.

Example :Pop-up advertisements

8. ROOTKIT

A Rootkit is malware designed to hide itself and provide attackers privileged access.

Exam Point

- Rootkits are difficult to detect.

RAT (REMOTE ACCESS TROJAN)

Full Form :Remote Access Trojan OR Remote Administration Tool

A RAT allows attackers to remotely control a victim's computer as if they were physically present.

Exam Point

- RAT provides complete remote control.

XSS ATTACK

Full Form:Cross-Site Scripting

An attack in which malicious scripts are injected into web pages.

Common Targets

- Comment Sections
- Feedback Forms
- Blogs
- Forums

Example

An attacker posts malicious JavaScript in a comment section.

Exam Point

- XSS = Cross-Site Scripting
- Targets website users.

SOFTWARE ATTACKS SUMMARY

Attack	Purpose
Virus	Infect Files
Worm	Spread Across Networks
Trojan	Disguise Malware
Spyware	Monitor Users
RAT	Remote Control
XSS	Inject Malicious Scripts
Ransomware	Encrypt Data for Money
Adware	Display Ads
Rootkit	Hide Malware

HARDWARE ATTACKS

Hardware attacks target physical components of computer systems.

TYPES OF HARDWARE ATTACKS

1. Manufacturing Backdoors

Hidden access points intentionally or unintentionally inserted during manufacturing.

2. Eavesdropping

Unauthorized listening or accessing protected information from hardware devices.

3. Fault Induction

Deliberately causing hardware faults to exploit system behavior.

4. Hardware Modification

Physically altering hardware components.

Example

Installing unauthorized chips.

5. Backdoor Creation

Creating secret methods to bypass authentication.

6. Counterfeiting

Creating fake hardware products to gain unauthorized access.

CYBER CRIME

Cyber Crime refers to criminal activities involving computers, networks, or digital devices.

Objectives

- Financial Gain
- Data Theft
- Identity Theft
- Espionage is the act of obtaining confidential, secret, or classified information without the permission of the person or organisation that holds it.
- Disruption

Examples

- Hacking
- Online Fraud
- Phishing
- Identity Theft
- Cyber Stalking

CYBER CRIMINAL

A person who commits cyber crimes.

Types

- Individual Hackers
- Organized Groups
- Terrorist Organizations
- Nation-State Attackers

CYBER TERRORISM

Cyber Terrorism refers to unlawful attacks against computers, networks, or information systems to create fear, disrupt services, or achieve political/social objectives.

Targets

- Government Systems
- Defense Networks
- Critical Infrastructure

Examples

- Attacking power grids
- Attacking military systems

CYBER ESPIONAGE

Cyber Espionage (Cyber Spying) is the unauthorized acquisition of confidential information for political, military, economic, or strategic advantage.

Targets

- Governments
- Defense Organizations
- Corporations

Purpose

- Collect Secrets
- Gather Intelligence
- Obtain Trade Secrets

Exam Point

- Cyber Espionage = Digital Spying

NIST CYBERSECURITY FRAMEWORK

Full Form:National Institute of Standards and Technology

Provides guidelines and best practices for cybersecurity risk management.

INFORMATION TECHNOLOGY ACT, 2000

The Information Technology Act, 2000 is India's primary law dealing with cyber crimes, electronic commerce, and digital signatures.

Exam Point

- IT Act came into force in 2000.

- India's first cyber law.

IMPORTANT IT ACT SECTIONS SUMMARY

Section	Related To
43	Damage to Computer System
66	Computer Related Offences
66B	Receiving Stolen Computer Resource
66C	Identity Theft
66D	Online Cheating by Personation

QUICK REVISION NOTES

Firsts

- Early experimental worm-like program = Creeper
- Early Creeper-removal program = Reaper

Malware

- Virus
- Worm
- Trojan
- Spyware
- Ransomware
- Rootkit
- Adware

Cyber Terms

- RAT = Remote Access Trojan
- XSS = Cross-Site Scripting
- NIST = National Institute of Standards and Technology

IT Act Sections

- 43 → Damage
- 66 → Computer-related offences involving dishonest or fraudulent acts listed in Section 43
- 66B → Dishonestly receiving a stolen computer resource or communication device
- 66C → Identity Theft
- 66D → Cheating by personation using a computer resource or communication device

MOST IMPORTANT FACTS FOR EXAMS

1.Creeper was the first virus-like program.

2.Reaper was the first antivirus.

3.RAT gives complete remote control over a device.

4.XSS stands for Cross-Site Scripting.

5.Worm spreads without a host file.

6.Trojan disguises itself as genuine software.

7.Spyware secretly collects user information.

8.Ransomware encrypts files and demands money.

9.Cyber Crime uses computers as a target or tool.

10.Cyber Espionage means digital spying.

11.Cyber Terrorism combines cyber attacks with terrorism.

12.NIST provides cybersecurity standards.

13.IT Act 2000 is India's primary cyber law.

14.Section 66C deals with identity theft.

15.Section 66D deals with online impersonation and cheating.

NETWORK SECURITY

Network Security refers to the policies, technologies, devices, and practices used to protect computer networks, data, and resources from unauthorized access, misuse, modification, or cyber attacks.

Objectives of Network Security

- Protect Data